Introduction to Apache Hive

Wednesday, 19 August 2026 8:46 PM

Q. What is Apache Hive?

It is a data warehouse tool. built on top of Hadoop.

Allows users to execute SQL style Queries.

Q. Do we need programming skills to work with Bigdata?

✓ MapReduce → Java.

HDFS: → Java.

Hive → SQL style Queries → Hive Query Language (HQL)

* Two Types of User Personas

1) MapReduce Users: Developers or Data Engineers.
More programming knowledge.

2) Hive User: Analysts, Business users } this will have more adoption.
SQL style queries.

* Differentiation between MapReduce & Hive

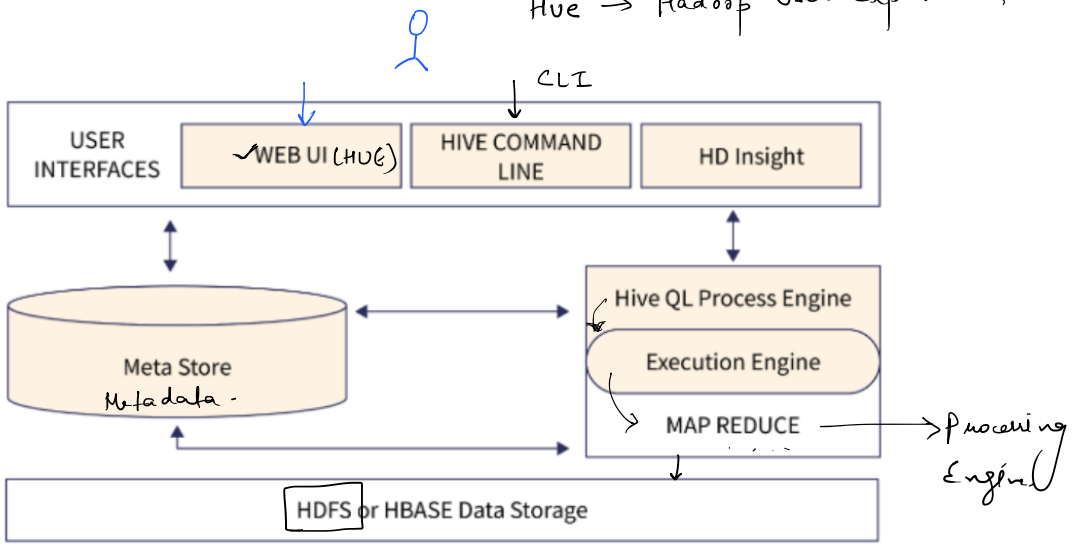
| Aspect | MapReduce | Hive |
|---|---|---|
| Basic Nature | Low-level data processing framework | High level processing framework |
| Programming | Requires programming (Java/Python) | No programming. SQL required |
| Target Users | Developers / Data Engineers | Analysts, Business Users. |
| Ease of learning | Difficult | Easy. |
| | less | More. |

| Abstraction. | | |
|---|---|---|
| Development time | High | low |
| Metadata Management | No centralized metadata | Uses Metastore. |
| Maintainance effort | High | Low |
| Flexibility | More flexible | less flexible. |

Q. How Hive Executes the queries – High level Flow.

1) User write a Hive Query (HQL)
2) Hive parses and validates the query
3) Query is converted into execution steps
4) Execution happens on Hadoop
5) Results are stored and returned to the user.

Hue → Hadoop User Experience.




Processing Engine

1) Hive command line Interface.
2) HUE (Hive User Experience)
3) JDBC → Java Database Connectivity
   ODBC → Open Database Connectivity
   } → allow access to external applications / BI tools.

✓ Important Components of Hive.

1) Driver:

→ Manages the lifecycle of the query.

performing parsing, compilation, optimization & execution.

→ Acts as a coordinator.

Coordinates interactions between hive components and the underlying execution engines.

2) Compiler:

→ Responsible for converting HiveQL into execution plan.

→ Syntax and semantic checks.

3) MetaStore:

It stores and manages all metadata information required by Hive.

database / tables

4) Hive Databases.

logical namespace used to organize and manage tables, views

5) Hive table:

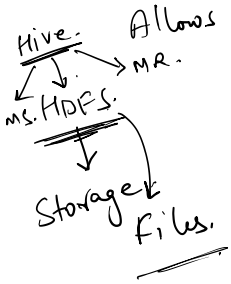
A structured view of the data stored in the underlying storage system. → HDFS.

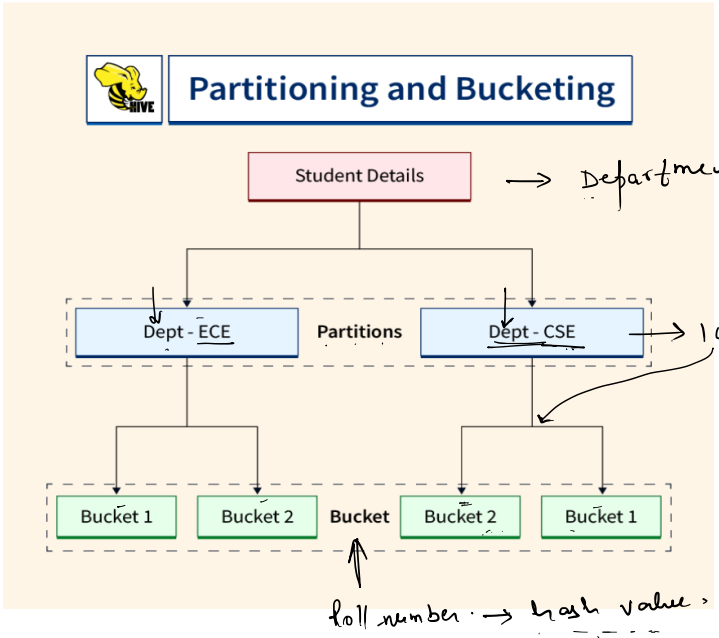
6) Partitions.

Divide data into smaller, manageable chunks. based on columns.

7) Buckets

Allows us to further divide data within table / partition.

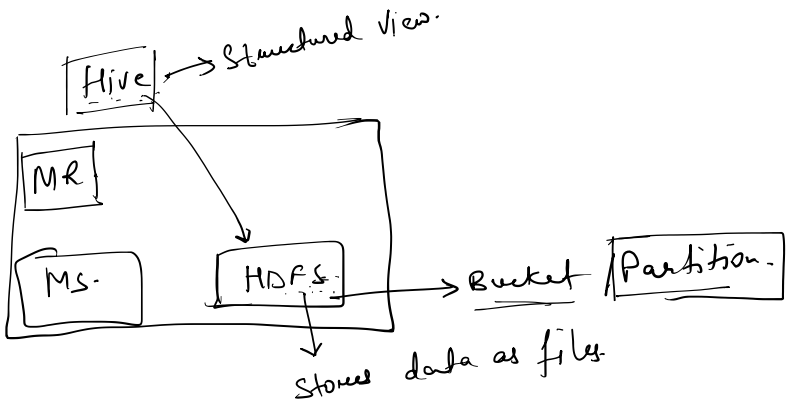



✳ Execution Engines.

1) MapReduce. → Hive 1.x, default is MR.

2) Tez / Spark. → Hive 2.x onwards, default is Tez.




Q. Can we select the execution engine in Hive?

SET hive.execution.engine = mr / tez / spark ;

* Hive Execution Engine Comparison:

| Aspect | MR (Legacy) | Tez (Modern Default) | Spark (Optional) |
|---|---|---|---|
| Execution model | Fixed Map-Reduce steps | DAG based execution | DAG Based execution. |
| Default in Hive | Hive 1.x | Hive 2.x | Not default, needs config |
| Performance | Slow | Faster | Fastest |
| DISK I/O | High | Medium to low | Minimal (in-memory processing) |
| Query latency | High | Medium to low | Low |
| Best suited for | Batch | Interactive & complex | Real time. |
| Resource utilization | less efficient | More efficient | Highly efficient. |